

Priyankkumar Chandrakant Patel

[linkedin.com/in/priyankk](https://www.linkedin.com/in/priyankk) | patel.priyankk@northeastern.edu | github.com/priyank1510 | (334) 343-2712

TECHNICAL SKILLS

Fundamental Concepts	:Object Oriented Designs, Algorithm, Data Structures, SDLC.
Programming Languages	:Java, Python, C, JavaScript, R, SQL, Typescript.
Databases and Cloud	:MySQL, MongoDB, SQLite, GCP, AWS, Spark, PostgreSQL.
Backend	:Spring Boot, FastAPI, Node.js, REST APIs, WebSockets, Agile.
Certifications	:AWS Cloud Architecting, AWS AML, Amazon Data Engineer.
Testing and Devops	:JUnit, Mockito, Git, GitHub, CI/CD, Docker.
AI/ML Framework	:Classification, Computer Vision, RAG, LLM, Clustering, Transformers, LangChain.

WORK EXPERIENCE

- AI Software Engineer Intern** | *Deepthink Healthcare* | San Ramon, USA Sep 2025 - Dec 2025
- Engineered a Python OCR preprocessing pipeline using Tesseract signals and Vision Transformer methods, improving the consistency of medical-document extraction and reducing manual review effort.
 - Evaluated **10,000+** healthcare question-answering responses through reproducible QA experiments, identifying **20** recurring failure patterns and contributing to **90%** reviewed accuracy.
 - Developed a transformer-embedding classifier for medical-device routing, achieving a **95%** F1 score and integrating model outputs into reusable Python evaluation pipelines.
 - Validated **5,000+** medical records in Label Studio against schema, completeness, and consistency rules, reducing annotation inconsistencies by **90%**.
- ML Data Scientist** | *Enlighten Infosystems* | Vadodara, India Jan 2023 - Sep 2023
- Analyzed **3,000+** sales records across five years using Python and SQL, transforming raw data into validated, model-ready datasets for business analysis.
 - Trained and benchmarked SVM and regression models, achieving **95%** classification accuracy and **92%** revenue-prediction accuracy on evaluation datasets.
 - Applied clustering algorithms to customer segmentation and communicated findings to business stakeholders, contributing to a **15%** improvement in targeted acquisition efforts.
- Software Development Engineer** | *Elsner Technologies* | Ahmedabad, India Apr 2022 - Aug 2022
- Delivered iOS task-management features with calendar integration, improving **user engagement by 30%**.
 - Optimized mobile app performance by **reducing load times 40%**, improving usability for large task lists.

PROJECTS

ML Production Readiness Toolkit

- Developed a reusable Python toolkit supporting **5+** Scikit-learn models and 15 configurable validation checks for schema, missing value, and feature-drift issues.
- Exposed standardized prediction, health-check, and monitoring endpoints through FastAPI, achieving sub 800 ms median latency across **2,000** test requests.
- Automated model packaging and Docker deployment, reducing API setup time from **2 hours** to under **20** minutes across **3** sample ML projects.
- Established a GitHub Actions CI pipeline with **40+** unit and integration tests, achieving **90%** test coverage across validation, model loading, and API workflows.

Real Time Event and Notification Platform

- Architected an event-driven notification platform using Java and Spring Boot to process order, payment, and account events and deliver real-time updates through WebSockets.
- Sustained **400+** simulated events per minute with sub 300 millisecond median API latency by using asynchronous processing, Redis caching, and indexed PostgreSQL queries.
- Verified delivery reliability through retry handling, event-status tracking, and 30+ JUnit and Mockito tests, achieving **80%** test coverage across critical workflows.

Clinical Evidence Graph RAG Platform

- Constructed a medical Graph RAG pipeline that extracted diseases, symptoms, medications, and treatments from **1,000+** PubMed abstracts and stored evidence linked relationships in **Neo4j**.
- Combined hybrid retrieval combining FAISS semantic search with multi-hop graph traversal to generate evidence grounded answers with traceable source citations.
- Benchmarked Graph RAG against vector only RAG across **500** PubMedQA questions, measuring answer accuracy, retrieval recall, citation precision, latency, and unsupported-claim rate.

EDUCATION

Master of Science in Computer Science Northeastern University, Boston, MA	Jan 2024 – May 2026 GPA: 3.8 / 4.0
Bachelor of Technology in Computer Science and Engineering Gujarat Technological University, India	Jun 2019 - May 2023 GPA: 4.0 / 4.0